

How AI Shapes the Research Agenda in Real Estate Finance and Economics

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Abstract

AI has begun to shape the research agenda in real estate finance and economics, not just accelerating research tasks. This paper maps the published literature into a semantic embedding space and uses it to evaluate 1,199 AI-generated research proposals across six prompting conditions, each replicated across four independent generation runs. Proposals are assessed using atypicality — a measure of cross-community recombination motivated by the science-of-science literature — validated against a citation-prediction model trained on the published corpus, which finds that realised citations are driven by venue and topic rather than by geometric position in the embedding space. The results show that AI can generate proposals comparable to published papers on the paper’s main novelty metric, but the outputs carry a persistent topical bias. Without corpus guidance, the model’s implicit priors over ‘real estate finance’ or ‘real estate economics’ tilt heavily toward urban-economics topics such as climate risk, rent control, and land use, while mortgage finance and institutional real estate are underrepresented; supplying the full published corpus does not override this prior, only supplementing it with a modest finance-specific tail. REE-corpus scaffolding produces the most atypical proposals, narrowly ahead of econ/finance method transfer; psychology-based transfers are narrower and, along with the other conditions, are frequently held back by data that does not clearly exist. The findings suggest that AI systems may increasingly influence not only how research is conducted, but which questions are attempted in the first place.

AI acknowledgement: AI was used extensively throughout this project. Literature discovery, data collection, empirical pipeline setup and econometric testing were all handled or substantially assisted by AI, mostly via Claude Code. Writing was facilitated with Claude Code and ChatGPT. I developed the research idea and empirical strategy, reviewed the literature, selected the papers, directed the AI agents on how to use them, audited intermediate results, redesigned parts of the workflow, checked the empirical setup and tests, and edited the final document. I did not write a single line of code, which feels strange for a former programmer. The history of my instructions

and prompts is available with all code at <https://github.com/thies/ai-idea-generation>. All errors are mine.

1 Introduction

Large language models (LLMs) are likely the fastest-adopted general-purpose technology in history¹. Within months of public release, systems such as ChatGPT, Claude and Gemini were being used across programming, education, law, medicine, finance and scientific research. Estimates of workplace adoption already exceed those observed for the internet or personal computers at comparable stages of diffusion (Bick et al., 2024; Chatterji et al., 2025), and the spread continues to accelerate.

Scientific research has not been exempt. Some commentators frame generative AI as a possible Kuhnian paradigm shift (Kuhn and Hacking, 1970) in scientific research. But is research actually changing at the level of scientific reasoning, or are we instead observing a subtler shift in how research agendas are formed and explored? Is generative AI simply another wave of task automation: a technology that allows researchers to do what they already did before, only faster and at lower cost?

One way to approach the distinction is to ask whether generative AI merely accelerates existing research workflows or whether it also influences what questions researchers choose to pursue. Can it generate ideas rather than simply execute them? Can it identify research gaps, combine methods across fields, or evaluate whether a result is plausible and important? In doing so, does it influence the direction of scientific discovery?

Even if generative AI cannot independently reason about science in a deep sense, this does not imply that research practices remain unchanged. Researchers will still probe, explore, experiment with, and misuse these systems. Some will embrace them enthusiastically; others will distrust or resist them. Either way, generative AI is already entering the research process itself. This is most clearly visible in our collective writing. Kobak et al. (2025), for instance, show that LLM-associated words rose sharply in biomedical abstracts after November 2022. Section 5.1 replicates this exercise for real estate and housing research and finds the same structural break. Our field is following the same trajectory as others in turning towards AI.

Beyond the writing layer, the more consequential shift is in using AI for more and more research tasks. A growing subset of publications in real estate economics and finance depends on AI not for drafting but for execution. Naik et al. (2016) use computer vision on street-level imagery to construct city-scale measures of urban appearance and link them to neighbourhood change. Lopez et al. (2022) extract structured information from MLS listings to document that agents representing foreclosure sellers steer buyers toward preferred lenders in over half of cases. Bartik et al. (2025) read and code thousands of municipal zoning ordinances. Shen and Ross (2021) extract a description-quality measure from MLS listing text. Leow and Lindenthal (2025) apply machine learning to REIT factor-return data, adapting the approach of Gu et al. (2020) in

¹Reuters: *ChatGPT sets record for fastest-growing user base - analyst note*, available at <https://www.reuters.com/technology/chatgpt-sets-record-fastest-growing-user-base-analyst-note-2023-02-01/>

stocks. In each case, AI enables a measurement or prediction that the paper requires. Remove it and the study does not exist.

The share of AI-empowered papers in the real estate literature is still small but rising fast. Searching the method and key-finding fields of 3,366 papers across *Real Estate Economics*, the *Journal of Real Estate Finance and Economics*, and the *Journal of Real Estate Research* for explicit AI/ML use yields 45 confirmed papers — 11 in REE, 21 in JREFE, and 13 in JRER. Isolated applications appear in JREFE and JRER as early as the 1990s, but the rate was low and sporadic until 2022. Since then, the annual count has risen to 5–6 papers across the corpus and the share of annual output approaches 6–8 per cent. This is a lower bound: it captures only papers where AI performs an explicit, otherwise infeasible task. A much larger grey zone covers data collection, screening, literature search, transcription and survey coding — uses that leave no methods-section trace. In the visible cases, AI is still a skilled research assistant (RA), executing tasks specified by a human principal investigator (PI).

Scholars increasingly use these systems not only for execution but also for ideation: brainstorming hypotheses, suggesting empirical designs, identifying literatures, proposing mechanisms, and reframing questions. Whether or not the underlying models genuinely “understand” research may therefore be less important than the fact that researchers behave as if they do.

This paper asks how far up the research value chain AI can go. Can it decide what is worth doing? The relevant cognitive step is to identify questions that are novel enough to matter and feasible enough to pursue. Si et al. (2025a) find that AI-generated ideas in NLP score higher on judged novelty than human ideas but lower on feasibility, and that they are less diverse than they appear. In a follow-up implementation study, the initial novelty advantage reversed: once researchers built the projects, human-suggested ideas were rated better (Si et al., 2025b).

But how should the quality of a research proposal be judged before the work exists? Researchers have been selecting topics for decades, and the published literature records the outcomes of those choices. As with asset prices, when theory alone cannot pin down value, observed market outcomes provide approximations. Here, I use the structure of the field’s published literature as an ex-ante proxy for scientific value. The intuition comes from Uzzi et al. (2013): highly cited papers often combine a conventional base with a small number of atypical knowledge combinations.

The coordinate system is built from the published archives of *Real Estate Economics*, the *Journal of Real Estate Finance and Economics*, real-estate-relevant subsets of the *Journal of Urban Economics*, and selected papers from leading economics and finance journals. A large language model extracts each paper’s research question, empirical method, data source, sector and main contribution. These structured representations are embedded in a shared semantic space and clustered into research strands. The result is measurement infrastructure rather than a conventional literature review: a map against which new proposals can be located relative to the field’s existing work. Because real estate research has repeatedly advanced by importing

ideas from neighbouring disciplines, the map is extended to selected papers in economics, finance and psychology to identify method–data–question combinations that are productive elsewhere but thinly represented in real estate economics.

In total, 1,199 research ideas are generated under six conditions and located in that space. The conditions vary in the context given to the model: no context, the full REE corpus, methods drawn from economics and finance, and methods drawn from psychology. Each idea is evaluated on atypicality, and a separate citation-prediction model calibrated on the published corpus tests whether geometric position in the embedding space actually predicts realised scientific impact. The analysis asks what different forms of scaffolding add over a naive prompt.

The results are encouraging but bounded. The best ideas, especially those generated through method transfer from economics and finance, are comparable to published REE papers on the paper’s main novelty metric. Some ideas simply rediscover questions the field has already answered. That is also true of human research proposals.

2 Related literature and research gap

2.1 AI in economics, finance, and adjacent fields

AI and ML have entered economics and finance as powerful research technologies. Athey and Imbens (2017) argue that ML methods hold promise for improving the credibility of causal identification and policy evaluation; their survey (Athey and Imbens, 2019) maps the tools empirical economists need to adopt, including regularisation, random forests, matrix completion and hybrid ML–econometric estimators. In finance, Gu et al. (2020) provide a benchmark illustration of what happens when machine learning is brought into an established empirical domain: improved predictive performance, new forms of non-linearity, and a clearer distinction between prediction and economic mechanism. Kelly and Xiu (2023) survey this financial machine learning literature more broadly, mapping the methods and open questions that remain. Bibliometric mapping of AI’s diffusion across economics more broadly, clustering thousands of articles by method and topic, confirms that the wave arrived first in prediction and macroeconomic modelling before spreading to more structural and theoretical applications (Bahoo et al., 2025).

The proliferation of AI-enabled systems raises new research questions, too. To offer just one example, improved prediction does not distribute its gains evenly. Fuster et al. (2022) show that switching to ML in US mortgage credit screening raises aggregate accuracy but reclassifies minority borrowers at systematically higher rates, establishing that distributional consequences are a first-order concern in high-stakes AI applications.

These literatures are relevant here for two reasons. AI typically enters a field first as a measurement or forecasting tool; and they offer mature method–data combinations that can be

compared against the real-estate research space for possible import.

2.2 Real estate finance and economics, machine learning, and AI as method

Within real estate finance and economics, the machine-learning literature has grown quickly, particularly around valuation and prediction. Tekouabou et al. (2023) survey AI-based methods for urban real-estate prediction and report systematic predictive gains relative to traditional hedonic approaches. Recent papers in REE and JREFE illustrate a broader shift. Lindenthal and Johnson (2021) use machine learning to classify architectural styles and link them to prices. Lorenz et al. (2023) develop interpretable machine learning for real estate market analysis. Leow and Lindenthal (2025) extend the Gu et al. (2020) machine-learning asset-pricing programme to REITs: applying ML to traditional factor-return data, they show substantial forecast improvements over OLS, a clean example of AI improving the analysis of conventional data without changing the underlying research question. Calainho et al. (2024) show the same pattern for commercial real estate price indices: replacing linear hedonic models with ML improves out-of-sample prediction accuracy on 30,000 New York transactions, while leaving the fundamental index construction problem unchanged.

A separate strand treats AI itself as the object of study: Wan and Lindenthal (2023) develop a framework for testing and validating machine learning systems in real estate. Beyond prediction, several papers demonstrate AI unlocking measurements that were previously impossible at scale, enabling traditional research questions to be answered with new data. Naik et al. (2016) use computer vision on street-level imagery to construct city-scale measures of urban appearance quality, applied to a classic question in urban economics about neighbourhood change. Shen and Ross (2021) apply unsupervised NLP to MLS listing descriptions, extracting a description-uniqueness measure that captures soft information about property quality, and find that a one-standard-deviation increase raises sale prices by around 15 per cent. Bartik et al. (2025) extend this logic to regulatory text: their generative regulatory measurement approach reads thousands of municipal zoning codes and extracts structured indicators at near-human accuracy, enabling a national analysis of housing regulation that would otherwise be infeasible.

These papers establish that AI adds value in real estate research. The gains are concentrated in measurement, prediction, feature extraction, and model support: RA tasks. But can AI do more?

2.3 Bibliometrics and the science of science

Donthu et al. (2021) survey bibliometric methods: citation analysis, co-authorship networks, and co-word mapping. Fortunato et al. (2018) situate such work within the science-of-science

programme, which studies the structure and dynamics of knowledge production itself.

The most directly relevant work concerns the relationship between novelty, combination, and impact. Uzzi et al. (2013) show that high-impact papers tend to combine a mostly conventional base of prior work with a small number of atypical combinations: pairings of knowledge that are unusual but not unprecedented. This empirical finding provides a bibliometric analogue to the incremental/recombinant/frontier distinction: high-impact work is not purely incremental (which lacks novelty) nor purely frontier (which lacks legibility), but sits at the boundary between the familiar and the new. Foster et al. (2015) extend this line of inquiry to research strategies, documenting how scientists trade off the higher expected rewards of exploratory research against the lower risk of incremental work, and showing that conservative strategies dominate even when novel research yields larger returns when it succeeds.

A second relevant strand uses semantic embeddings to characterise research spaces rather than citation networks. Tshitoyan et al. (2019) demonstrate that word embeddings trained on materials science abstracts encode implicit knowledge about element combinations and can predict future discoveries by identifying concepts that are close in embedding space but not yet combined in the literature. This is a direct precursor to the approach taken here: the embedding space is not merely a visualisation of what has been done, but a structure that encodes what is possible at the frontier.

This paper differs in focusing on research content rather than metadata, and in using the embedding space as an evaluation device for new proposals rather than a description of existing ones.

2.4 LLM use in academic writing, scientific discovery, and idea generation

A parallel literature documents how researchers themselves use AI tools. Kobak et al. (2025) track words that language models overuse relative to human writers to detect LLM involvement in scientific abstracts. Their method provides a reproducible lower bound: authors who edit LLM drafts carefully leave no trace. Section 5.1 replicates the analysis for the real-estate literature.

A growing literature asks whether LLMs can contribute to scientific discovery beyond summarisation and coding. Si et al. (2025a) provide the closest benchmark: a large-scale human evaluation of AI-generated NLP ideas, finding higher judged novelty but lower feasibility. Their follow-up implementation study found an ideation-execution gap: ideas that looked stronger at the proposal stage were rated worse after execution, while human-generated ideas improved (Si et al., 2025b). This is a direct challenge to any evaluation method that stops at ideation. In the natural sciences, execution can sometimes be validated experimentally. In social science and economics, no equivalent bench test exists. This asymmetry motivates the approach taken here: using the structure of the field’s existing literature, and specifically atypicality, as a measurable

ex-ante proxy for scientific value. Baek (2025) develop an iterative agent that queries the literature to refine proposals; Su et al. (2025) show that structured debate among multiple LLM agents improves idea quality over single-model generation.

More recent agentic systems push beyond idea generation: Gottweis et al. (2025) describe Google’s AI co-scientist, which generates and debate-evolves hypotheses with experimentally validated results in biomedicine; Lu et al. (2026) automate the full pipeline from idea to manuscript, identifying evaluation as the binding constraint. Beyond ideation, recent systems target evaluation (Goyal et al., 2026) and closed-loop experimental reasoning (Cissé et al., 2026), where the binding constraint shifts from generating hypotheses to verifying them.

2.5 Novelty, research strategy, and the frontier

Uzzi et al. (2013) establish that the most-cited papers tend to combine conventional and atypical knowledge in specific proportions: genuine advances are rarely purely incremental or purely sui generis. Foster et al. (2015) show that scientists systematically underinvest in novel research relative to the social optimum, preferring incremental strategies that reduce the risk of failure. Wang et al. (2017) document that truly novel papers are initially under-cited and face systematic resistance from peer review, suggesting that the research space is not a level playing field: ideas that are genuinely distant from the current frontier face higher barriers to publication and recognition.

These findings shape how to interpret the proposal-location results. A proposal in a sparse or frontier region is not guaranteed to be valuable, it may simply be unfamiliar. But the science-of-science evidence suggests the expected payoff, conditional on acceptance, is higher than for incremental work. Embedding distance is therefore informative about risk and return, not just novelty.

Tshitoyan et al. (2019) add a further dimension: the embedding space trained on a corpus encodes not only what has been done but what the literature is ready to support. Concepts that co-occur frequently but have not yet been combined in a single paper represent low-hanging combinations: the background knowledge exists, but synthesis is still missing. The most productive frontier proposals are therefore not those at maximum distance from existing work but those bridging dense clusters: recombinant rather than purely novel.

2.6 Methods imports and cross-field transfer

Real estate finance and economics has repeatedly advanced by importing methods from neighbouring literatures. Hedonic pricing comes from environmental and labour economics; event studies from finance; regression discontinuity and modern causal inference from applied microeconomics. This history motivates the cross-field transfer exercise. Rather than asking only what is missing within real estate, the exercise asks which combinations of methods, data, and questions

are already productive in other fields, finance, economics, psychology, and allied data-rich social sciences, but have not yet been systematically brought into real estate economics.

At the same time, not every transfer is valuable. Prediction methods transplanted into causal settings without regard to identification, institutional context, or the data-generating process do not necessarily advance economic knowledge. The purpose of the cross-field comparison is therefore not simply to import novelty, but to identify imports that satisfy the inferential standards of the receiving field.

3 Data

Gentzkow et al. (2019) provide the canonical overview of text as data in economics. The use of large language models here extends this tradition: structured summaries are extracted from each paper’s full text, covering research question, method, data source, sector, and main finding. This identifies substantive similarities between papers that use different surface vocabularies, and permits direct comparison along dimensions that matter for economic research, not only what topic is studied, but how, with what data, and to what inferential end.

The empirics rely on core real estate journals. *Real Estate Economics (REE)*, formerly the *Journal of the American Real Estate and Urban Economics Association*, is the field’s oldest specialist journal, published continuously since 1973. Metadata for all 1,676 articles were collected via CrossRef; full-text PDFs were obtained for 909 articles from 1997–2026 via the Wiley Text and Data Mining API under institutional licence. Pre-1997 papers are not available through this channel.

The *Journal of Real Estate Finance and Economics (JREFE)* has been published by Springer since 1988 and covers real estate finance, investment and urban economics. Metadata for 1,693 articles were collected via OpenAlex; full text is available for 250 articles via the Springer Open Access API (JATS XML) and open-access PDFs, with structured LLM extraction applied to all 1,693 entries.² The *Journal of Real Estate Research (JREER)*, published continuously since 1986 by the American Real Estate Society, covers applied real estate analysis and practice. Metadata for 1,171 articles were collected via OpenAlex, with full-text extraction from abstracts. The *Journal of Urban Economics (JUE)* is broader, but is a major outlet for housing supply, land use and spatial economics. I retain only papers whose OpenAlex topics include housing markets, land use, urban development, valuation or closely related areas.

A targeted subset of real estate, housing, mortgage, and urban papers is drawn from leading general-interest journals: the *Journal of Finance*, *Review of Financial Studies*, and *American Economic Review*. The supplementary sample is not intended to provide exhaustive coverage of those fields. Its purpose is to support cross-field transfer analysis by identifying method–data–

²<https://dev.springernature.com/>

question combinations that are productive elsewhere but thinly represented in real estate finance and economics.

The psychology component draws on four journals in the judgment, decision-making, and organisational-behaviour tradition: *Judgment and Decision Making* (JDM), *Journal of Behavioral Decision Making* (JBDM), *Journal of Economic Psychology* (JEP), and *Organizational Behavior and Human Decision Processes* (OBHDP). This is a specific behavioural-decision corpus, not a representative sample of psychology as a discipline: it does not cover social, clinical, developmental, or cognitive-perceptual psychology, and the transfer results in Section 6.2 should be read as evidence about transfer from judgment-and-decision-making research specifically, not from “psychology” in general.

Citation counts, reference lists, concept tags, and journal metadata are obtained via OpenAlex.³ Where full text is unavailable, title, abstract, and reference information are retained for cross-field comparisons.

The RE corpus — REE, JREFE, and JRER — defines the field’s research space and provides the base case for the idea-generation experiment. Together they cover core real estate finance and economics from 1973 to the present. The economics and finance journals (JUE, AER, JF, RFS) supply the cross-field transfer pool; the psychology journals serve the same role for behavioural methods and questions.

4 Method

For each paper with a full-text PDF, the text is extracted and an LLM queried⁴ with a structured prompt⁵ to recover six fields: `research_question`, `method`, `data`, `sector`, `ai_ml_role` and `key_finding`.

The substantive extraction fields are concatenated into a single text representation per paper and encoded using `all-mpnet-base-v2`⁶, generating one embedding per paper. All journals are embedded in a shared space so that distances are directly comparable across outlets and fields.

³OpenAlex is a free, fully open bibliometric database maintained by OurResearch (<https://openalex.org>), covering over 250 million scholarly works with metadata, abstracts, citation networks, and concept tags.

⁴All extractions use `claude-sonnet-4-6` (Anthropic). This may need revisiting if model capabilities or instruction-following behaviour diverge substantially in future generations.

⁵The system prompt used for all extractions reads in full: “*You are a research assistant analysing academic real estate economics papers. Given information about a paper, extract exactly five fields as JSON: `research_question`, one sentence: the main research question or objective; `method`, one sentence: the primary empirical or theoretical method; `data`, one sentence: the main data source(s) and coverage; `key_finding`, one sentence: the main result or contribution; `ai_ml_role`, one sentence: the role of AI/ML in this paper, or ‘None’ if absent. Be concise. Use plain English. Do not copy abstract text verbatim.*” The user message appends the paper title, abstract, and a full-text excerpt (up to approximately 8,000 tokens) where the PDF is available, or title and abstract only where it is not.

⁶<https://huggingface.co/sentence-transformers/all-mpnet-base-v2>

Dimensionality reduction is used for visualisation and clustering, with clustering performed in a higher-dimensional reduced space rather than in two-dimensional plots. HDBSCAN identifies dense regions and leaves low-density points unassigned instead of forcing every paper into a cluster. The 1,884 unassigned papers (30.1%) mark genuinely diffuse territory. Cluster labels are generated from representative papers and refined by human inspection. Dense clusters mark likely incremental ground; sparse bridging regions hold recombinant combinations; frontier regions contain configurations largely absent from the existing literature.

Research proposals are generated with different contexts, prompts and cross-field transfers, then embedded into the shared space. Each proposal is expressed in a structured form: research question, empirical design, likely data source, expected contribution and relevance to real estate economics. Incremental proposals fall inside dense clusters; recombinant proposals lie between clusters; frontier proposals lie at or beyond the observed support of the literature.

This measures distance from existing research configurations, not innovation itself. Distance does not imply importance, correctness or feasibility, but it provides a replicable way to compare whether ideas are more of the same or genuinely different. The question is descriptive: where do proposals from different sources land, and does scaffolding change that distribution?

Papers from the *Journal of Finance*, *Review of Financial Studies*, and *American Economic Review* are embedded alongside the specialist-journal corpus and their cluster positions compared. The comparison is descriptive: publication venue is not a clean quality signal, since real estate papers in top journals are drawn from the intersection of quality and legibility to a general-audience editor, not from the full frontier of the field. Journal type is treated as one covariate alongside citation counts, year, and topic.

For cross-field transfer, comparable embeddings are constructed for selected papers in finance, economics, psychology, and related disciplines, and clusters or local configurations that are productive there but underrepresented in real estate economics are identified. The key safeguard is economic validity: imported methods are only counted as promising when the underlying inferential logic matches the target real-estate question. A purely predictive architecture imported into a causal policy setting does not qualify.

5 Results

5.1 LLM use in real estate research writing

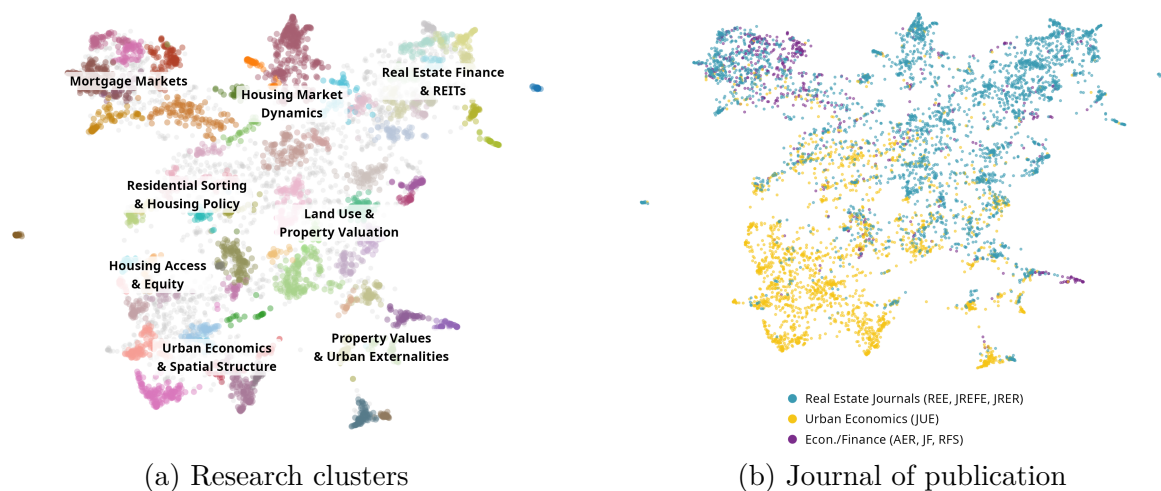
Kobak et al. (2025) identified words that large language models overuse relative to human writers — “delve,” “pivotal,” “nuanced,” “comprehensive,” “crucial” — and showed their frequency in biomedical abstracts jumped after ChatGPT’s November 2022 launch. The same method is applied to two real-estate corpora: the *Real Estate Economics* archive, too small for year-level inference, and roughly 101,000 OpenAlex-indexed real-estate, housing, and urban-economics

papers (2010–2025), which provides the quantification (Figure 4, Appendix A). Aggregate indicator-word rates were stable through 2022 at roughly 0.85 per 1,000 abstract words, then tripled to 2.45 by 2024; “delve” alone rose twenty-fold, replicating Kobak et al. (2025) in a field with no prior documentation of LLM-assisted writing. Excess-word rates are a lower bound — careful editing leaves no signature — and some flagged words have legitimate technical uses excluded from the trend; conversely, Yakura et al. (2025) show frequent LLM users reproduce AI-preferred words in their own speech, so not every occurrence is a fingerprint of AI-written prose.

5.2 The research space of real estate economics

The combined embedding space contains 6,255 papers and is partitioned into 78 clusters. Of these, 92 are administrative entries — issue information, book reviews, and editorial board listings — that embed as extreme outliers in the UMAP projection; they are excluded from Figure 1’s journal-group breakdown, leaving 6,163 papers shown there. The full cluster table appears in the appendix. Each cluster is labelled by a short topic description generated by an LLM from a sample of paper titles and research questions. Figure 1 shows the research space coloured by cluster and by journal.

Figure 1: Research Space of Real Estate Finance and Economics



Notes: All papers embedded using the same sentence transformer; 6,163 of the full 6,255-paper corpus are shown, excluding 92 administrative entries (issue information, book reviews, editorial board listings) that appear as extreme outliers in the UMAP projection. (a) Coloured by HDBSCAN cluster; meta-cluster labels positioned at cluster centroids. (b) Coloured by journal of publication: Real Estate Journals, $n = 3,341$; Urban Economics, $n = 2,219$; Econ./Finance, $n = 603$. Each point is one paper; position reflects semantic similarity of research question, method, and context.

5.3 The emergence of AI

AI and machine learning papers within the RE corpus are identified by searching the LLM-extracted fields for method-specific terms (machine learning, neural networks, random forests, gradient boosting, NLP, computer vision, word embeddings, textual analysis). Full-text extraction catches roughly a quarter more papers than abstract-only search in the JREFE/JRER subsample where both exist; the approach only captures explicit AI use for otherwise-infeasible tasks, not AI used invisibly for data collection or screening.

This yields 45 AI/ML papers: 11 in REE, 21 in JREFE, 13 in JRER. Isolated applications — mostly early neural-network property valuation — appear in JREFE and JRER as early as the 1990s, but adoption was low and sporadic; REE has none before 2020. The first concentration emerges in 2022–2023 (6 papers each year), reaching 5–6 per year by 2024–2025.

Adoption is heavily concentrated by topic: 24 of the 45 papers (53 per cent) belong to a single cluster, automated property valuation — the natural entry point, since ML improves nonparametric price-function estimation without requiring model specification and the data are abundant and structured. The remaining 21 span 10 clusters (market liquidity, mortgage default, spatial econometrics, REIT forecasting, transit capitalisation, rent control, residential brokerage) plus 6 in the unclustered noise class; AI papers do not form a methodological island, they sit embedded within existing topic clusters where the methods happen to be useful (Figure 2).

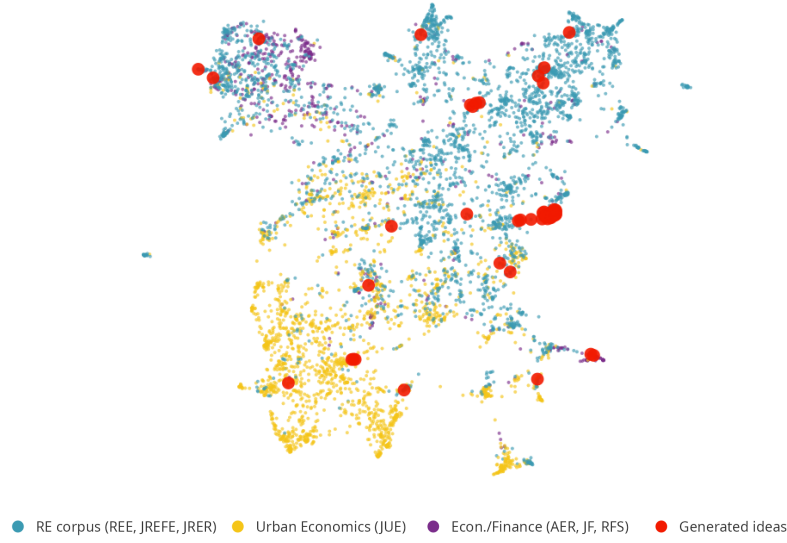
Most papers use AI as a *measurement or prediction tool*: Lorenz et al. (2023) compare XGBoost against a conventional hedonic specification for residential rents, improving precision while retaining interpretability; Francke and van de Minne (2024) apply a random-effects/ML hybrid to commercial real estate in Phoenix, reaching sub-11-per-cent out-of-sample error. Lopez et al. (2022)’s foreclosure-steering finding (Section 1) is the exception: ML as a text-mining device making an institutional fact observable, not a forecasting tool.

None of the 45 papers uses AI for causal-inference support or studies AI as an economic phenomenon — the two roles most prevalent in the neighbouring economics and finance journals. The RE corpus’s AI/ML papers sit uniformly at the measurement-and-prediction end of the spectrum: a tool for doing something the researcher already specified, not a co-designer of the research itself.

5.4 Real estate papers in economics and finance journals

The combined embedding space includes 3,366 papers from the RE corpus (REE, JREFE, JRER) and 2,889 papers drawn from the top economics and finance journals (JUE, AER, JF, RFS). Because all papers are embedded using the same LLM-extracted structured representations and the same sentence transformer, the resulting clusters reflect content similarity rather than source. A paper that asks the same question using the same method as a REE article will land in the same neighbourhood regardless of where it was published.

Figure 2: AI/ML Papers in the Research Space



Notes: Background points coloured by journal group (see legend); $n = 6,163$ papers with $\text{umap_x} < 15$. Red dots = AI/ML-classified papers across all journals in the embedding space ($n = 54$: 11 REE, 21 JREFE, 13 JRER, 6 JUE, 3 econ/fin); the 45 RE corpus papers discussed in Section 5.3 are the REE, JREFE, and JRER subset. Each point is one paper; position reflects semantic similarity in the shared embedding space. AI/ML papers are identified by keyword search of LLM-extracted method and key-finding fields.

Of the 78 clusters, 64 contain papers from both source types; 8 contain only top-journal papers; 6 contain only RE corpus papers. The research space of REE and its sister journals therefore overlaps substantially with the broader real-estate and economics literature, with pockets of RE-specific and top-journal-specific territory.

The 8 exclusively top-journal clusters, dominated by JUE, cover congestion pricing, firm location and agglomeration, spatial labour markets, climate risk and financial markets, urban structure and growth, city size distribution, and urban/spatial equilibrium theory — active areas of urban economics and finance that never fed back into specialist real estate journals.

Among the 64 mixed clusters, the RE corpus and top-journal papers frequently cover the same territory. Topics established first in general economics — land use regulation, local tax policy, racial segregation in housing — were subsequently picked up by the field journals.

Within the RE corpus, cluster composition tracks citations: RE papers in RE-dominated clusters (under 25 per cent top-journal peers) are cited markedly more than those in top-journal-dominated clusters (over 75 per cent peers), and even after age adjustment, RE papers in mixed clusters have significantly lower citation residuals than their top-journal neighbours — consistent with the citation advantage of higher-ranked outlets, though possibly also reflecting self-selection at submission.

The six RE-only clusters are small and concentrated in niche applied topics; the eight

top-journal-only clusters cover active areas of urban economics not yet represented in specialist real estate journals. The cross-field transfer analysis in Section 6 therefore asks which regions of the broader space are densely populated elsewhere but thinly represented in the RE corpus, and whether AI-assisted method transfer can help close the gap.

6 AI as Principal Investigator: Generating and Evaluating Research Proposals

The preceding sections document AI extracting information, classifying documents, and measuring the structure of the literature. This section asks whether AI can act as a principal investigator: generating proposals that are not merely plausible-sounding but meaningfully novel.

6.1 Experimental design

Ideas are generated under six conditions using the same large language model. Each condition is run twice: once unconstrained and once with a hard data-existence requirement embedded in the prompt (“only propose designs where you can confirm the required data exists and is accessible to an academic researcher”). This yields twelve condition-variants, each generated as four independent runs (forty-eight runs in total), and allows me to test whether the constraint changes what gets proposed or merely removes ideas that a post-hoc feasibility assessment would have flagged.

Condition A is a naive prompt with no corpus context. Condition B supplies 779 extracted paper summaries from REE. Conditions C and D add the full pool of methodologically distinctive papers from economics/finance (612 papers: AER, JF, RFS) or psychology (1,714 papers: JDM, JBDM, JEP, OBHDP) and ask the model to transfer methods into real estate research, while preserving the inferential logic of the target field. Conditions E and F go further by transplanting both the research question and empirical design from the source field. Each condition-variant pools four independently seeded runs of 25 ideas: each run draws its own random ordering of the source material and carries no memory of the other three, so no idea is conditioned on what a prior run already proposed.

All ideas share a common output structure: research question, empirical method, required data, expected contribution, self-rated feasibility and novelty rationale. Claimed novelty is extracted from the rationale in a separate call and compared with each idea’s geometric position in the embedding space. The scaffold is the treatment; the embedding pipeline, atypicality measure and citation-prediction regression are held constant.

6.2 Evaluation and results

Each run produces 100 ideas, each scored on atypicality and feasibility and assigned to the nearest HDBSCAN cluster. Atypicality is the Shannon entropy of the cluster distribution among an idea’s 20 nearest neighbours, normalised to $[0, 1]$: it captures whether an idea bridges several research communities rather than extending a single one, which differs from nearest-neighbour distance (how far an idea sits from existing papers) — an idea can be close to existing work but combine several strands, or distant from all existing work while extending only one. Because the citation model already penalises raw distance from existing work, atypicality is treated as the main ex-ante signal of productive recombination. Feasibility is handled first through the generation-stage data-existence constraint, then checked in Section 6.3 through a structured post-hoc assessment. Table 1 summarises means pooled across each condition-variant’s four independent runs; Figure 3 shows where ideas land relative to the existing corpus. Four independent runs per cell is a real improvement over a single run — no source-paper subsample and no cross-run conditioning can drive an apparent condition effect — but it remains a small number of runs for formal hierarchical inference, so the comparisons below should still be read as descriptive rather than as confirmatory population-level tests.

Figure 3 maps where each run’s ideas land in the embedding space; the Herfindahl–Hirschman index (HHI) in Table 1 makes the concentration precise. The RE corpus baseline HHI is 0.030. Naive generation and the non-psychology conditions cluster in a similar range (A: 0.051–0.065; B: 0.044–0.053; C: 0.059–0.065; E: 0.060–0.073), reflecting that neither an REE-only corpus nor economic and financial transfer methods funnel ideas into a narrow slice of the space. Psychology-sourced conditions are markedly more concentrated (D: 0.113–0.130; F: 0.145–0.156, roughly four to five times the baseline): the model gravitates to a narrow behavioural territory — loss aversion in selling, anchoring in appraisals, attention effects in search — and the data constraint reduces this only slightly.

The naive prompt reveals the model’s prior over “real estate.” Without corpus context, Condition A’s top clusters are natural disasters and climate risk (13%), rent control policy (8.5%), land use regulation (8.5%), and mortgage lending discrimination (7.5%) — all squarely urban-economics-adjacent, despite the prompt explicitly asking for ideas across “housing markets, commercial real estate, mortgage markets, land use, REITs, climate risk.” Supplying the full REE corpus (Condition B) does not override this prior: the same four topics still dominate (climate risk 10%, rent control 9.5%, mortgage lending discrimination 7%, land use 5%). What the corpus adds is a modest but real finance-specific tail that is close to absent without it — REIT market microstructure, portfolio optimisation, market liquidity and efficiency, mortgage securitisation, and interest-rate/credit cycles each go from roughly 0–1 per cent of ideas under A to 1–3.5 per cent under B. Corpus scaffolding supplements the model’s dominant prior rather than displacing it.

Weak data support is not confined to the psychology-sourced conditions. The share of ideas

Table 1: Research idea quality and spatial distribution by generation condition

Condition	N	Atypic.	Clust.	HHI	Weak data	Periph.	NN dist.
A: Naïve	100	0.726 ± 0.214	36/78	0.051	10%	3.52	1.14
— <i>constr.</i>	100	0.715 ± 0.227	29/78	0.065	3%	3.17	1.10
B: Full REE corpus	100	0.772 ± 0.173	35/78	0.044	27%	4.17	1.16
— <i>constr.</i>	100	0.770 ± 0.158	38/78	0.053	12%	3.64	1.12
C: Method trans. (econ/fin)	100	0.745 ± 0.192	30/78	0.065	9%	2.87	0.93
— <i>constr.</i>	100	0.713 ± 0.225	31/78	0.059	5%	3.23	0.96
D: Method trans. (psych)	100	0.673 ± 0.238	27/78	0.130	35%	2.65	1.09
— <i>constr.</i>	100	0.661 ± 0.270	29/78	0.113	19%	3.06	1.09
E: Paradigm trans. (econ/fin)	99	0.729 ± 0.238	29/78	0.073	15%	2.89	0.99
— <i>constr.</i>	100	0.755 ± 0.181	28/78	0.060	4%	3.10	0.85
F: Paradigm trans. (psych)	100	0.682 ± 0.258	31/78	0.145	42%	3.34	1.17
— <i>constr.</i>	100	0.674 ± 0.275	25/78	0.156	25%	3.11	1.03
RE corpus (baseline)	2374	0.594 ± 0.258	70/78	0.030	—	1.00	1.00

Notes: Each condition appears twice: unconstrained and constrained (marked *constr.*); the constrained variant requires the model to confirm data availability before proposing a design. Each condition-variant pools four independently seeded runs of 25 ideas, drawing on the full source pool (no source-paper subsampling). Atypicality = mean $\pm$ SD; Shannon entropy of the 20-nearest-neighbour cluster distribution, normalised to $[0, 1]$. The RE corpus baseline uses the same measure computed leave-one-out (each paper’s own 20 nearest neighbours, excluding itself); its $N = 2,374$ is the cluster-assigned subset of the full 3,366-paper RE corpus (REE, JREFE, JRER), excluding the 992 papers HDBSCAN leaves unassigned. Cluster coverage = number of the 78 named HDBSCAN clusters receiving at least one assigned idea; noise papers (cluster -1) are included as a 79th category in the HHI calculation. HHI = Herfindahl–Hirschman index of cluster shares; lower values indicate ideas spread more evenly across the research space. Weak data = share of ideas whose structured post-hoc assessment (Section 6.3) rated data existence 0–1 on a 0–3 scale, i.e. the required data has weak precedent or does not clearly exist. Periph. / RE = within-cluster peripherality (Euclidean distance from cluster centroid in UMAP space) normalised so the RE corpus mean = 1.00; values above 1 indicate ideas land farther from cluster centres than the typical corpus paper. NN dist. / RE = mean cosine nearest-neighbour distance in the sentence embedding space, normalised to the RE corpus mean; values above 1 indicate ideas are more distant from existing work than the average RE corpus paper.

Figure 3: Generated Ideas in the Research Space: All 12 Generation Runs (part 1 of 2)

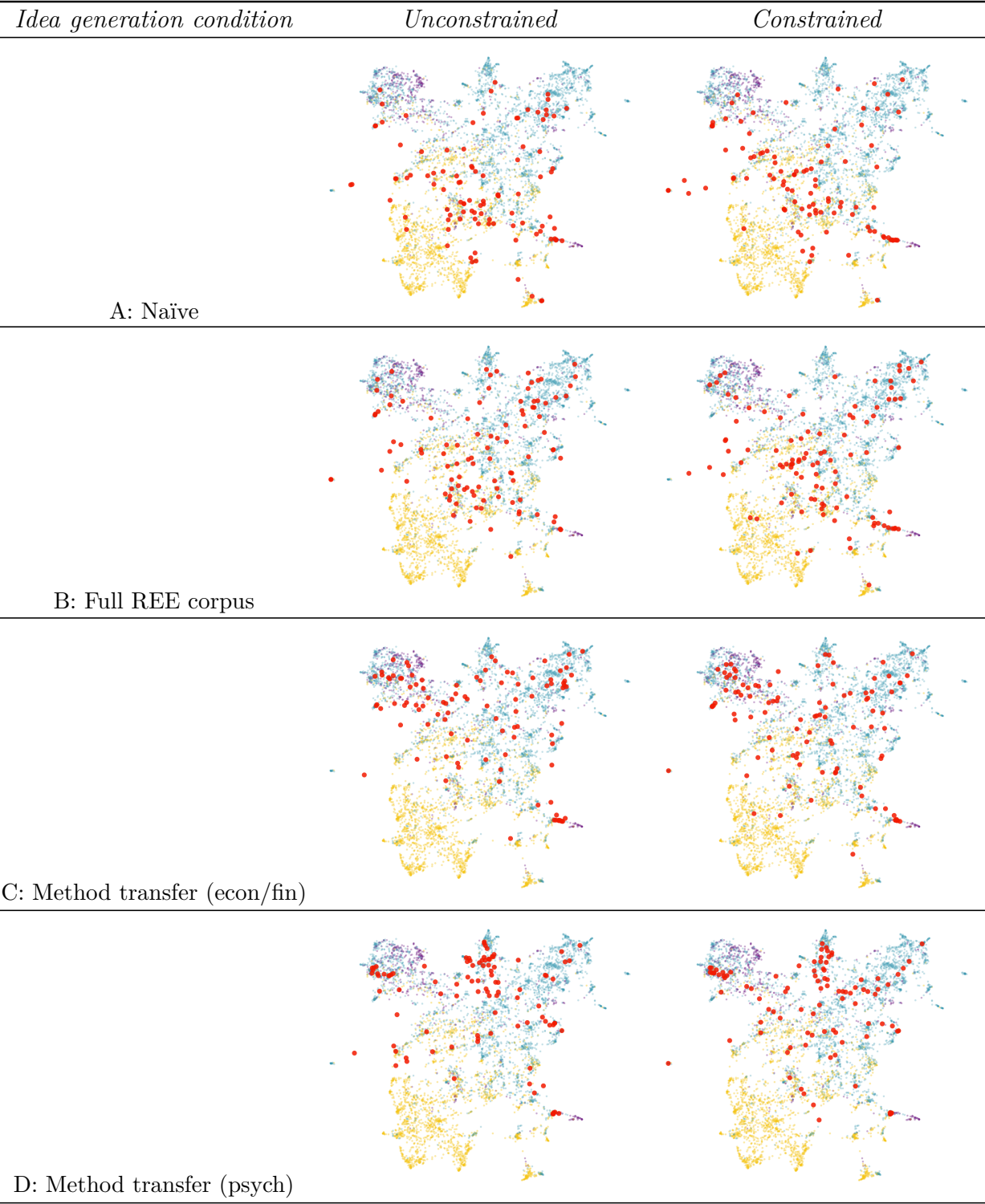
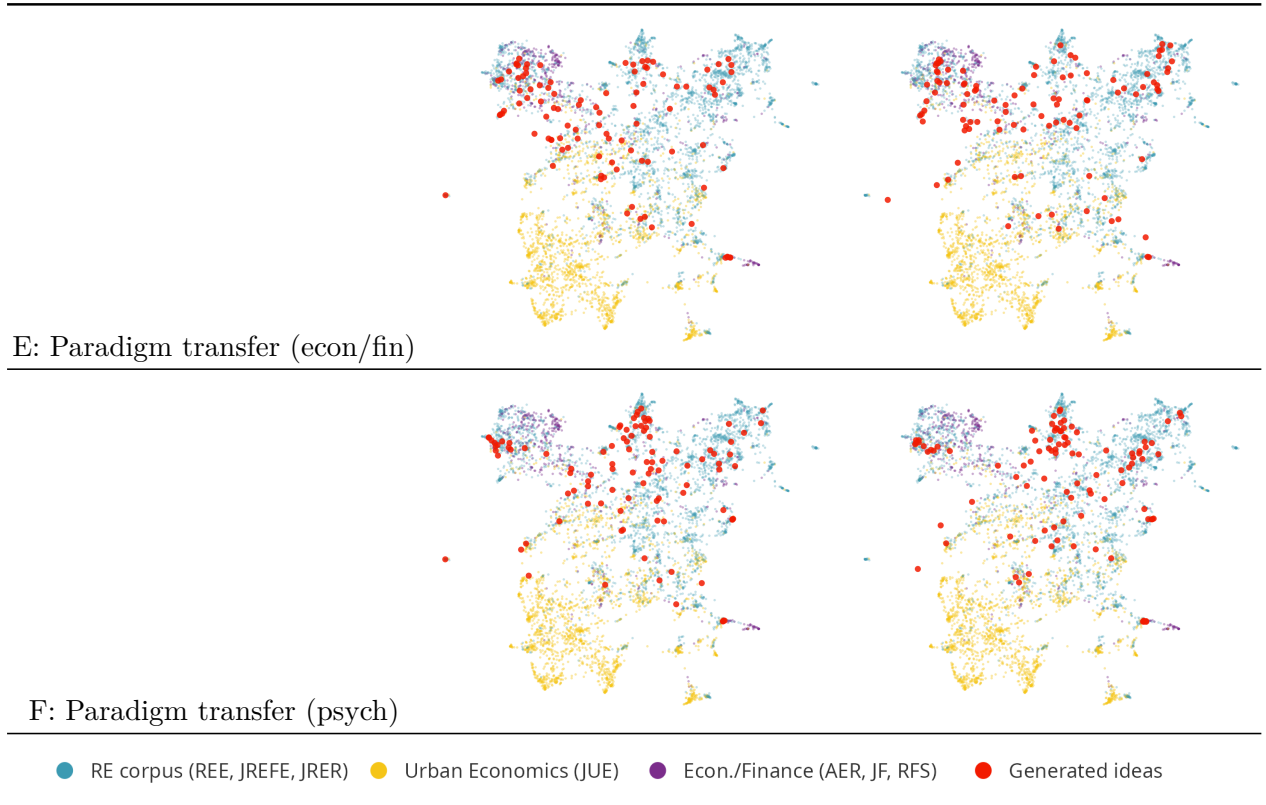


Figure 3 (continued)



Notes: Each panel projects generated ideas (red) into the UMAP space fitted on the full corpus. Background points are coloured by journal group (see legend). Ideas are embedded using the same sentence transformer and projected via the pre-fitted UMAP model. Unconstrained variants allow the model to generate freely; constrained variants include the instruction “only propose designs where you can confirm the required data exists and is accessible to an academic researcher.” The five book-review entries in the far right of the corpus are excluded from the background. An interactive version of this figure is available at <https://thies.github.io/idea-explorer/>.

whose required data the structured assessment rates 0–1 out of 3 (Table 1, “Weak data”) is highest for psychology paradigm transfer (F: 42% unconstrained, 25% constrained) and psychology method transfer (D: 35%, 19%), but Condition B is a distant third at 27% unconstrained — higher than either econ/finance condition (C: 9%; E: 15%) and higher than the naive prompt itself (A: 10%). A large REE context does not guarantee that the ideas it inspires are grounded in accessible data; it can also prompt the model to reach for institutionally plausible but practically unavailable sources.

Spatial position tells a different story from topic concentration. Every condition is markedly peripheral relative to the corpus: mean distance from cluster centroids (Periph/RE) ranges from 2.65 (D, unconstrained) to 4.17 (B, unconstrained), two-and-a-half to four times the typical published paper’s distance from its own cluster centre. Condition B — generated with no source-field anchor beyond the REE corpus itself — is the most peripheral of all six conditions, not the least, reversing what a naive reading of its high atypicality might suggest. Nearest-neighbour distance in the embedding space tells a calmer story: at 0.85–1.17 times the RE corpus mean, generated ideas sit about as close to their single nearest existing paper as a typical corpus paper does. Together the two measures describe ideas that are not unusually distant from any one precedent, but that consistently land at the edges of the topics they belong to rather than at their core.

The model is also a poor judge of its own novelty: of the 1,199 ideas, 88.9 per cent receive a claimed novelty score of 3 out of 4 and only 0.2 per cent score 4, yet claimed novelty correlates just 0.012 ($p = 0.69$) with actual atypicality. More context does not improve calibration.

The ANOVA on atypicality across the six conditions is significant ($F = 6.18$, $p < 0.001$), driven mainly by the psychology-sourced conditions trailing the rest; among the others, Condition B leads (0.771 pooled), narrowly ahead of econ/finance paradigm transfer (E: 0.742) and method transfer (C: 0.729), with the naive prompt close behind (A: 0.721).

Atypicality is motivated as an ex-ante signal of productive recombination, but that motivation rests on an empirical claim: that geometric position in the embedding space — how atypical, how distant from existing work, how peripheral within a cluster — actually predicts a paper’s realised scientific impact. I test this directly on the published corpus rather than assume it, since generated ideas have no citation history of their own to check it against. I train an OLS model of $\log(1 + \text{citations})$ on 1,937 comparable papers published 2001–2021, with HC3-robust standard errors, regressing on $\log(\text{age})$, atypicality, nearest-neighbour distance, within-cluster peripherality (log frontier distance), cluster mean log-citations, and journal fixed effects. Geometric predictors are computed against the historically available corpus at each paper’s publication year to avoid look-ahead bias. In-sample $R^2 = 0.374$, Spearman $r = 0.601$; 5-fold cross-validated $R^2 = 0.362$, Spearman $r = 0.595$, confirming negligible overfitting.

The model estimated is:

$$\begin{aligned} \log(1 + c_i) = & \alpha + \beta_1 \log(\text{age}_i) + \beta_2 \text{atypicality}_i + \beta_3 \text{nn_dist}_i \\ & + \beta_4 \log(\text{frontier_dist}_i) + \beta_5 \bar{c}_{k(i)} + \sum_j \gamma_j \mathbf{1}[\text{journal}_i = j] + \varepsilon_i \end{aligned} \quad (1)$$

Summary statistics for the training set are reported in Table 3 (Appendix B).

Table 2: OLS citation prediction model

	Coefficient	SE
log(Age)	0.747***	0.053
Atypicality	−0.162	0.113
NN distance	−0.194	0.392
log(Frontier dist)	−0.078***	0.030
Cluster mean log-cit	0.679***	0.035
AER	1.634***	0.214
JF	1.970***	0.240
JREFE	−0.531***	0.060
JRER	−1.041***	0.079
JUE	0.606***	0.085
RFS	1.555***	0.167
Constant	−0.829***	0.216
N		1,937
R ²		0.374

Notes: OLS with HC3-robust standard errors. Reference category: REE. Geometric predictors computed against the historically available corpus at each paper’s publication year to avoid look-ahead bias. $p < 0.01$, $p < 0.05$, $p < 0.10$.

The dominant predictors are log(age), nearest-neighbour distance and journal fixed effects. Atypicality is not a significant predictor of realised citations once cluster citation norms and proximity are controlled, and its point estimate is negative. That is not evidence against atypicality as a research-quality signal: Wang et al. (2017) document that genuinely novel papers are systematically under-cited relative to incremental work, at least in the citation windows typically available to researchers, so a null or negative coefficient is what their account predicts rather than a contradiction of it. The honest reading of this regression is a negative result: none of the geometric measures — atypicality, nearest-neighbour distance, peripherality — robustly predicts realised impact in the published corpus once venue and cluster-level citation norms are controlled. Citations are driven overwhelmingly by where a paper is published and what its neighbours already earn, not by its position in the embedding space. This is a reason to treat citation counts as uninformative about the merit of an unpublished proposal, and to rely on atypicality as a measurable ex-ante property whose relationship to eventual scientific value remains, on the evidence gathered here, unvalidated rather than confirmed.

6.3 Structured feasibility assessment

To test the generation-stage data constraint, a structured external assessment rates each idea 0–3 on data precedent, data existence, method–data fit and question plausibility; the composite score is the mean of the four divided by 3. Self-rated and structured feasibility correlate moderately ($r = 0.442$, $p < 0.001$) — a real signal, but not close enough to substitute for the external check.

Non-psychology conditions score higher on average (A, B, C, E: 0.733) than psychology conditions (D, F: 0.689; ANOVA across all six conditions $F = 14.19$, $p < 0.001$), because imported psychological methods rely on lab infrastructure or survey instruments with no real estate counterpart. Data existence is the binding constraint across every condition (1.68–2.03 out of 3), consistently below question plausibility (2.52–2.94): the model constructs plausible mechanisms more reliably than it proposes data that actually exists.

The constrained prompt does not fix this. It roughly halves the share of ideas flagged for weak data support (F: 42% to 25%; D: 35% to 19%, Table 1) but moves the overall structured composite only slightly (0.706 unconstrained to 0.731 constrained) — the constraint changes what the model *says* about its data requirements more than what it actually proposes. A post-hoc structured assessment remains the only reliable filter.

7 Conclusion

AI in academic research is like microplastics: pervasive, hard to trace, and accumulating quietly. Whether this amounts to a Kuhnian paradigm shift is probably the wrong question. The writing evidence shows AI-associated language already tripled in two years across the real estate literature; ideation is likely to follow the same curve. The Condition A results make the risk concrete: without corpus scaffolding, the model’s prior over “real estate economics” gravitates toward climate risk, rent control, and land use regulation — while mortgage finance, capital markets, and institutional real estate barely register (Figure 3). Supplying the model with the full REE corpus does not override this prior; it only adds a modest, real supplement of finance-specific topics on top of the same dominant agenda. If a growing share of research proposals is filtered through systems with that prior, the field’s agenda will drift — not because anyone decided it should, but because the model’s training distribution has its own centre of gravity, and that centre of gravity resists correction by context alone. A baked-in prior is not inherently a problem: Ross et al. (2026) show that purposefully designed LLM conversations can durably correct financial misconceptions that traditional literacy interventions struggle to shift. But that correction depends on corrective intent — the same paper finds that undirected conversations can actively entrench misconceptions instead of fixing them. Both Condition A and Condition B are undirected in exactly this sense: neither instructs the model to correct anything, and what both entrench is not the correction of an error but a partial view of a field that is in fact broad. Mortgage finance, capital markets and institutional real estate are real

estate economics too, and merely handing the model more real estate papers does not by itself register them. The same mechanism has been documented outside research ideation: Tsirtsis et al. (2026) show that LLMs mediating human-to-human communication introduce directional biases which, propagated through a social network, shift collective opinion at equilibrium. A fixed prior over what “real estate economics” is works the same way on a research field — not by persuading anyone of anything, but by quietly narrowing what gets proposed. Because scientific attention is path dependent, even small shifts in AI-assisted ideation may compound over time into larger shifts in the field’s research agenda. The result is not a revolution in how science is done but a slow bias on what gets proposed, attempted and published.

For now, AI adoption in published real estate research remains concentrated in measurement and prediction tasks: AI as research assistant rather than principal investigator. Yet the more consequential development may lie less in task execution than in the gradual integration of AI into the exploratory stages of research itself.

The idea-generation experiment produces 1,199 proposals under six conditions. REE-corpus scaffolding produces the most atypical ideas, narrowly ahead of the two econ/finance transfer conditions. Citations turn out to be the wrong yardstick for judging unpublished proposals: in the published corpus, none of the geometric measures — atypicality, nearest-neighbour distance, peripherality — robustly predicts realised citations once journal and cluster norms are controlled, and where atypicality does have a sign it is negative, consistent with the finding that genuinely novel work is initially under-cited. Citations, in other words, are venue- and topic-driven and therefore uninformative about the merit of an idea that has not yet been tested by peer review. Atypicality remains the paper’s measurable ex-ante property; whether it tracks eventual scientific value is, on the evidence gathered here, an open question rather than a settled one. The model’s confidence in its own proposals carries a real but partial signal about executability ($r = 0.442$) — not enough to substitute for external checking; data existence is the binding constraint across every condition, not only the psychology-sourced ones, and a generation-stage data constraint changes what the model says about data more than what it actually proposes.

Some ideas do not convince. Many rediscover questions the field has already answered. But the best, especially those generated through econ/finance method transfer, are comparable to published REE papers on the paper’s main novelty metric. A few are genuinely striking: original combinations with plausible mechanisms and accessible data that, to the best of my knowledge, have not been attempted. The boundary between research assistant and principal investigator is narrowing but not disappearing. The main remaining constraint is not fluency or even novelty. It is executability. This echoes Si et al. (2025a)’s finding that AI-generated ideas score higher on novelty than feasibility, and Si et al. (2025b)’s ideation–execution gap, where proposals that looked strongest before execution fared worst after it: the binding constraint on AI-generated research is not what the model can imagine, but what a researcher can actually build.

The underlying models are also advancing quickly: a result documented against one snapshot

may be stale within months and should be re-examined as new systems are released. All generation here uses a single general-purpose model (`claude-sonnet-4-6`); a different model, or one purpose-built for ideation, may behave differently, and nothing here establishes that the naive prior documented in Condition A generalises across model families. Finally, a sceptic could argue that this naive-prompt bias is less consequential than it looks, since a careful user could steer the model toward the full breadth of the field with a better prompt — I am not persuaded, since the naive prompt already explicitly asks for ideas across “housing markets, commercial real estate, mortgage markets, land use, REITs, climate risk,” and the model still defaults to urban economics regardless.

Data and code availability

All code will be made available at the <https://github.com/thies/ai-idea-generation>. Structured extraction outputs, embeddings, and cluster assignments can be shared subject to publisher licensing constraints. Raw PDFs cannot be redistributed due to copyright limitations.

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References

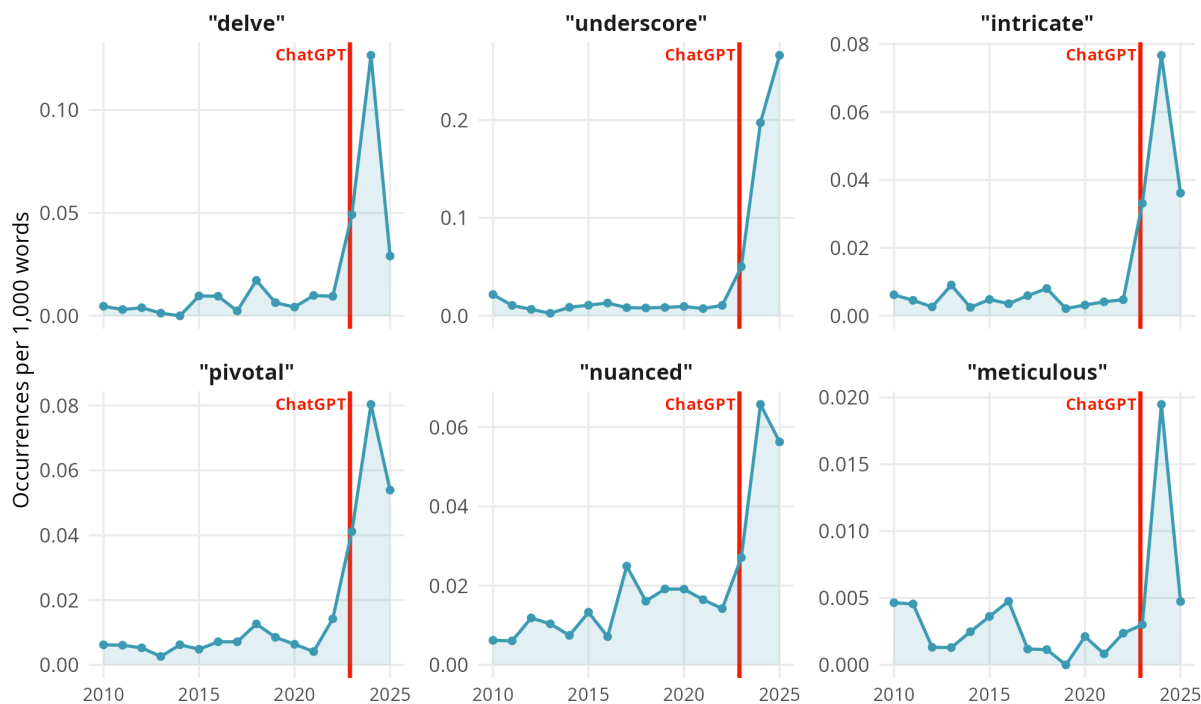
- Athey, S. and G. W. Imbens, “The State of Applied Econometrics: Causality and Policy Evaluation,” *Journal of Economic Perspectives*, 2017, 31 (2), 3–32.
- and —, “Machine Learning Methods That Economists Should Know About,” *Annual Review of Economics*, 2019, 11, 685–725.
- Baek, J., “ResearchAgent: Iterative Research Idea Generation over Scientific Literature,” 2025.
- Bahoo, S., J. Cuñado, and K. Gupta, “Artificial Intelligence in Economics Research: What Have We Learned? What Do We Need to Learn?,” *Journal of Economic Surveys*, 2025.
- Bartik, A. W., A. Gupta, and D. Milo, “The Costs of Housing Regulation: Evidence from Generative Regulatory Measurement,” 2025. Working paper.

- Bick, Alexander, Adam Blandin, and David J. Deming, “The Rapid Adoption of Generative AI,” Working Paper 32966, National Bureau of Economic Research September 2024.
- Calainho, F. D., A. M. van de Minne, and M. K. Francke, “A Machine Learning Approach to Price Indices: Applications in Commercial Real Estate,” *Journal of Real Estate Finance and Economics*, 2024, 68, 624–653.
- Chatterji, Aaron, Thomas Cunningham, David J. Deming, Zoe Hitzig, Christopher Ong, Carl Yan Shan, and Kevin Wadman, “How People Use ChatGPT,” Working Paper 34255, National Bureau of Economic Research September 2025.
- Cissé, A., M. E. Cooper, M. Zhu, X. Evangelopoulos, and A. I. Cooper, “Can We Automate Scientific Reasoning in Closed-Loop Experiments using Large Language Models?,” 2026.
- Donthu, N., S. Kumar, D. Mukherjee, N. Pandey, and W. M. Lim, “How to conduct a bibliometric analysis: An overview and guidelines,” *Journal of Business Research*, 2021, 133, 285–296.
- Fortunato, S., C. T. Bergstrom, K. Börner, J. A. Evans, D. Helbing, S. Milojević, and A.-L. Barabási, “Science of science,” *Science*, 2018, 359 (6379), eaao0185.
- Foster, J. G., A. Rzhetsky, and J. A. Evans, “Tradition and Innovation in Scientists’ Research Strategies,” *American Sociological Review*, 2015, 80 (5), 875–908.
- Francke, M. and A. van de Minne, “Combining machine learning and econometrics: Application to commercial real estate prices,” *Real Estate Economics*, 2024, 52, 1308–1339.
- Fuster, A., P. Goldsmith-Pinkham, T. Ramadorai, and A. Walther, “Predictably Unequal? The Effects of Machine Learning on Credit Markets,” *Journal of Finance*, 2022, 77 (1), 5–47.
- Gentzkow, M., B. Kelly, and M. Taddy, “Text as Data,” *Journal of Economic Literature*, 2019, 57 (3), 535–574.
- Gottweis, Juraj, Wei-Hung Weng, Alexander Daryin, Tao Tu, Anil Palepu, Petar Sirkovic, Artiom Myaskovsky, Felix Weissenberger, Keran Rong, Ryutaro Tanno et al., “Towards an AI co-scientist,” 2025.
- Goyal, P., M. Parmar, Y. Song, H. Palangi, T. Pfister, and J. Yoon, “ScholarPeer: A Context-Aware Multi-Agent Framework for Automated Peer Review,” 2026.
- Gu, S., B. Kelly, and D. Xiu, “Empirical Asset Pricing via Machine Learning,” *Review of Financial Studies*, 2020, 33 (5), 2223–2273.
- Kelly, B. and D. Xiu, “Financial Machine Learning,” 2023. Working paper.
- Kobak, D., R. González-Márquez, E.-Á. Horvát, and J. Lause, “Delving into ChatGPT usage in academic writing through excess word analysis,” *Science Advances*, 2025, 11 (3), eadt3813.
- Kuhn, Thomas S. and Ian Hacking, *The structure of scientific revolutions*, Vol. 2, University of Chicago press Chicago, 1970.
- Leow, K. and T. Lindenthal, “Enhancing Real Estate Investment Trust Return Forecasts using Machine Learning,” *Real Estate Economics*, 2025.

- Lindenthal, T. and E. B. Johnson, “Machine Learning, Architectural Styles and Property Values,” *Journal of Real Estate Finance and Economics*, 2021.
- Lopez, L. A., S. J. McCoy, and V. Sah, “Steering consumers to lenders in residential real estate markets,” *Real Estate Economics*, 2022, *50* (4), 1596–1641.
- Lorenz, F., J. Willwersch, M. Cajias, and F. Fuerst, “Interpretable Machine Learning for Real Estate Market Analysis,” *Real Estate Economics*, 2023, *51* (7), 1689–1719.
- Lu, C., C. Lu, R. T. Lange, Y. Yamada, S. Hu, J. Foerster, D. Ha, and J. Clune, “Towards end-to-end automation of AI research,” *Nature*, 2026.
- Naik, N., R. Raskar, and C. A. Hidalgo, “Cities Are Physical Too: Using Computer Vision to Measure the Quality and Impact of Urban Appearance,” *American Economic Review*, 2016, *106* (5), 128–132.
- Ross, Jillian, Eric So, and Andrew W. Lo, “Breaking Bad Financial Habits: How LLM Conversations Correct Financial Misconceptions,” 2026.
- Shen, L. and S. L. Ross, “Information value of property description: A machine learning approach,” *Journal of Urban Economics*, 2021, *121*, 103299.
- Si, C., T. Hashimoto, and D. Yang, “Can LLMs Generate Novel Research Ideas? A Large-Scale Human Study with 100+ NLP Researchers,” in “ICLR 2025” 2025a.
- , —, and —, “The Ideation–Execution Gap in AI-Generated Research Ideas,” 2025b.
- Su, H., R. Chen, S. Tang, X. Yin, J. Zheng, B. Qi, Q. Wu, H. Li, W. Ouyang, P. Torr, B. Zhou, and N. Dong, “Many heads are better than one: Improved scientific idea generation by a LLM-based multi-agent system,” *ACL 2025*, 2025.
- Tekouabou, S. C. K., Ş. C. Gherghina, E. D. Kamani, Y. Filali, and K. Idrissi Gartoumi, “AI-Based Machine Learning Methods for Urban Real Estate Prediction: A Systematic Survey,” *Archives of Computational Methods in Engineering*, 2023, *31*, 1079–1095.
- Tshitoyan, V., J. Dagdelen, L. Weston, K. A. Persson, G. Ceder, and A. Jain, “Unsupervised word embeddings capture latent knowledge from materials science literature,” *Nature*, 2019, *571*, 95–98.
- Tsirsis, Stratis, Kai Rawal, Chris Russell, Brent Mittelstadt, and Sandra Wachter, “AI-Mediated Communication Can Steer Collective Opinion,” 2026.
- Uzzi, B., S. Mukherjee, M. Stringer, and B. Jones, “Atypical Combinations and Scientific Impact,” *Science*, 2013, *342* (6157), 468–472.
- Wan, W. X. and T. Lindenthal, “Testing Machine Learning Systems in Real Estate,” *Real Estate Economics*, 2023.
- Wang, J., R. Veugelers, and P. Stephan, “Bias against novelty in science: A cautionary tale for users of bibliometric indicators,” *Research Policy*, 2017, *46* (8), 1416–1436.
- Yakura, H., E. Lopez-Lopez, L. Brinkmann, I. Serna, P. Gupta, I. Soraperra, and I. Rahwan, “Empirical evidence of Large Language Model’s influence on human spoken communication,” 2025.

A Supplementary figures

Figure 4: LLM-Indicator Word Rates in Real Estate and Housing Research, 2010–2025



Notes: Annual rate (occurrences per 1,000 abstract words) of fifteen LLM-associated words identified by Kobak et al. (2025) in approximately 101,000 papers indexed by OpenAlex under real-estate, housing, urban economics, and REIT concepts. Rates are aggregated across all indicator words. The dashed vertical line marks November 2022 (ChatGPT launch).

B Citation-prediction training set

C Generation prompt templates

This appendix reports the exact prompt templates used for each generation condition. All API calls use model `claude-sonnet-4-6` with the system prompt below. Each condition-variant is generated as four independent runs of 25 ideas each; every run is a single, self-contained API call with its own random seed and its own shuffled ordering of the source material, and carries no list of ideas from other runs to avoid repeating.

System prompt (all conditions)

You are an expert in real estate economics and finance. Generate precise,

Table 3: Summary statistics: citation prediction training set (N = 1,937)

Variable	Mean	SD	Min	Median	Max
log(1 + citations)	3.13	1.44	0.00	3.22	7.82
Citations (raw)	58.9	127.2	0	24	2,484
log(Age)	2.58	0.48	1.61	2.71	3.22
Atypicality	0.683	0.225	0.000	0.745	1.000
NN distance	0.237	0.078	0.000	0.233	0.558
Frontier distance	0.247	0.205	0.002	0.207	3.464
log(Frontier dist.)	-1.712	0.874	-6.454	-1.576	1.242
Cluster mean log-cit.	3.184	0.836	0.000	3.171	4.752
AER (dummy)	0.034	0.181	0	0	1
JF (dummy)	0.014	0.117	0	0	1
JREFE (dummy)	0.360	0.480	0	0	1
JRER (dummy)	0.164	0.370	0	0	1
JUE (dummy)	0.201	0.401	0	0	1
RFS (dummy)	0.033	0.177	0	0	1

Notes: Training set for the OLS citation-prediction model of Section 6.2 (Table 2). 1,937 papers from REE (377), JREFE (698), JRER (317), JUE (389), AER (66), RFS (63), and JF (27) published 2001–2021 with valid embeddings, cluster assignment, and citation count; an initial pool of 2,367 candidate papers is reduced to 1,937 after dropping papers with missing frontier-distance or cluster-mean-citation features. Age measured as of 2026. Geometric features (atypicality, NN distance, frontier distance) computed against the historically available corpus at each paper’s publication year. Journal dummies are the regressors entering Eq. 1 (REE is the omitted reference category, at 19.4% of the sample); frontier distance is shown both raw and log-transformed since the regression uses the log form.

original, and impactful academic research ideas. Aim for ideas that could become seminal papers -- genuine advances in method, question, or data, not incremental extensions of standard approaches.

Return a JSON array. Each object must have exactly these keys:

"research_question": one sentence stating the main research question,
"method": one sentence describing the primary empirical or theoretical method,
"data": one sentence describing the main data source(s) and coverage,
"key_finding": one sentence describing the plausible range of outcomes,
"feasibility": integer 0-10 (10 = highly feasible),
"novelty_rationale": one sentence explaining what makes this idea novel.

Constrained variant (additional clause, all conditions)

The following sentence is appended to the user prompt in every constrained run:

IMPORTANT: Only propose designs where the required data already exists and is accessible to an academic researcher (publicly available datasets,

administrative data accessible via research agreements, or commercial data used in published real estate research). Do not propose designs requiring data that does not yet exist or is practically inaccessible.

Condition A — Naïve

No context is provided. User prompt:

Generate {n} original real estate economics and finance research ideas. Cover the full breadth of the field: housing markets, commercial real estate, urban economics, mortgage markets, land use, REITs, climate risk, discrimination, neighbourhood effects, and so on.

Condition B — Full REE corpus

The user message has two content blocks. The first (cache-eligible) block contains 779 REE paper summaries, shuffled independently for each run:

=== Real Estate Economics Research Corpus ===

Title: [title, up to 100 characters]

RQ: [research_question, up to 200 characters]

Method: [method, up to 150 characters]

Finding: [key_finding, up to 150 characters]

[repeated for all 779 REE papers, independently shuffled per run]

The second (uncached) block:

Generate {n} original research ideas that extend or complement the corpus above. Each idea should be clearly distinct from the papers shown.

Conditions C and D — Method transfer (econ/fin and psych)

The cached block contains the REE research corpus (779 papers), structured as:

=== Real Estate Economics Research Corpus ===

Title: [title, up to 100 characters]

RQ: [research_question, up to 200 characters]

Method: [method, up to 150 characters]

Finding: [key_finding, up to 150 characters]

[repeated for all 779 REE papers, independently shuffled per run]

The uncached block appends the source-field papers, then the generation instruction. For Condition C the source header is `=== Economics and Finance Source Papers ===` (the full pool of 612 AER, JF, and RFS articles with a usable research question; JF, the largest of the three, supplies 345). For Condition D it is `=== Psychology and Behavioural Science Source Papers ===` (the full pool of 1,714 papers across JDM, JBDM, JEP, and OBHDP). In both cases:

`=== [Field] Source Papers ===`

These papers represent methods and approaches rarely used in real estate economics. Use them as inspiration for transferable designs.

Title: [title]

RQ: [research_question]

Method: [method]

Finding: [key_finding]

[repeated for the full source-field pool, independently shuffled per run]

Generate research ideas that apply methods from the source-field papers to questions in real estate economics -- but only where the inferential logic of the method transfers cleanly to the real estate setting.

Generate {n} ideas.

Conditions E and F — Paradigm transfer (econ/fin and psych)

Same cached REE corpus and source papers as Conditions C and D respectively. Only the generation instruction changes:

Generate research ideas that transplant both the research question and empirical design from the source-field papers to real estate contexts, testing whether the same phenomena or mechanisms operate in property markets, housing decisions, or real estate finance.

Generate {n} ideas.